DATA 110 · PROJECT 1 · FALL 2026

Does spending more per student mean more college graduates?

Scope of Analysis

Maryland's 23 counties and Baltimore City

Presented By

Matthew Velasquez

Montgomery College

THE DATASET

Choose Maryland: Compare Counties, Education

24 rows

23 counties plus Baltimore City, one row each

10 columns

County, graduates, student-teacher ratio, spending per pupil, bachelor's %, high school %, and 2-year and 4-year college counts and enrollment

Missing values

Blank enrollments mean the county has no such college. Dorchester has 1 four-year college and no enrollment listed

THE QUESTION

Do counties that spend more per student, have more college-educated adults?

The data

24 Maryland jurisdictions, one row each

The measures

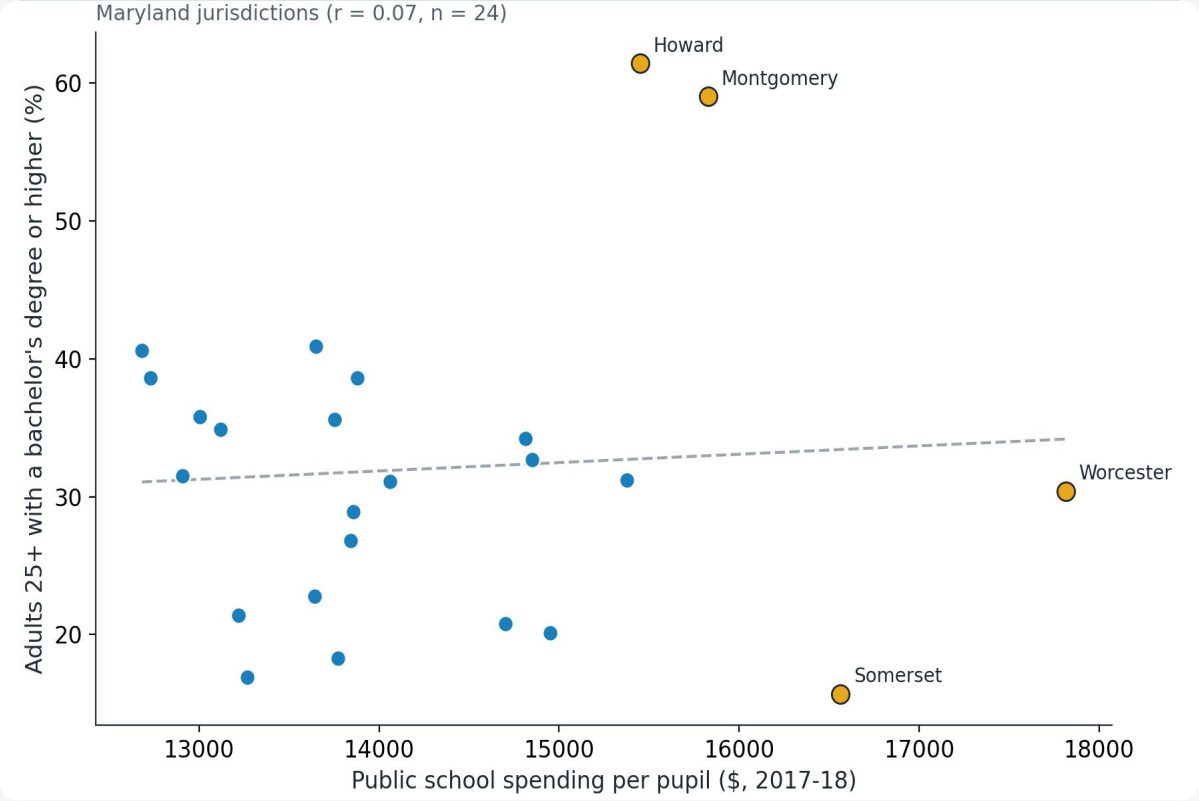
Public school spending per pupil, and the share of adults 25+ with a bachelor's degree

The sources

Maryland State Department of Education and the U.S. Census Bureau, 2014–2018

CHART 1 · SCATTER

Spending and degrees are almost unrelated



$\bar{r} = 0.07$

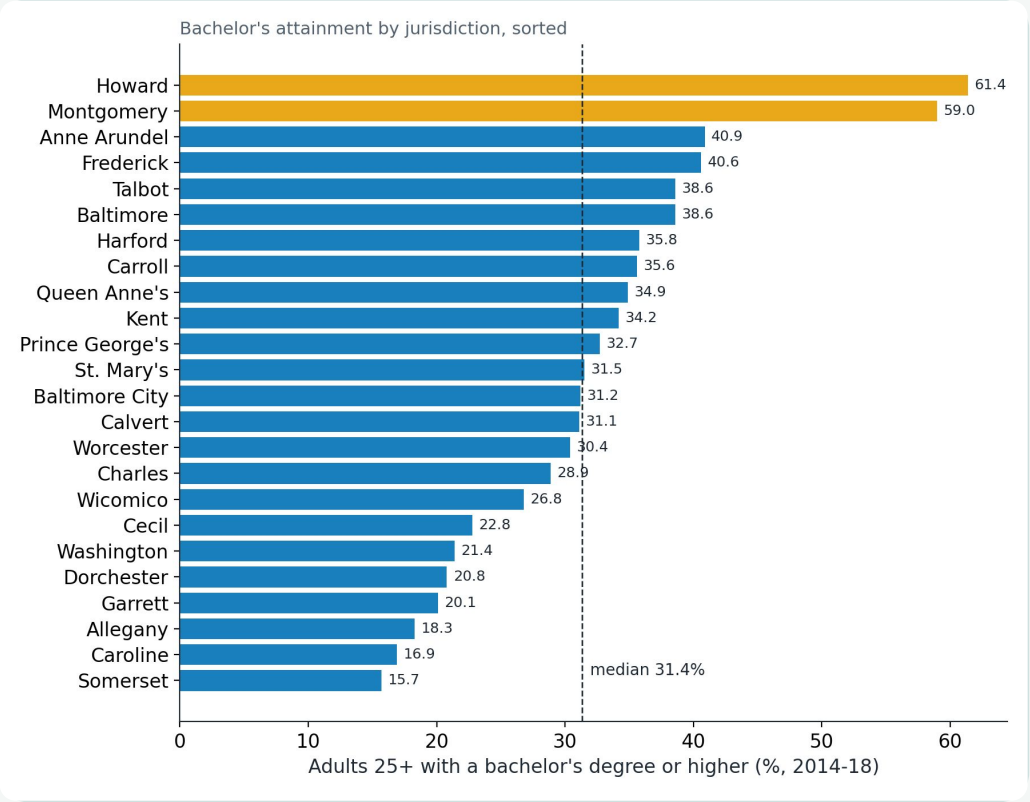
spending vs. bachelor's %

Somerset spends $16,563 per pupil and has 15.7% bachelor's attainment. Howard spends $15,449 and has 61.4%.

Source: Maryland State Department of Education; U.S. Census Bureau

CHART 2 · BAR

Two counties are far ahead of the rest



61.4%

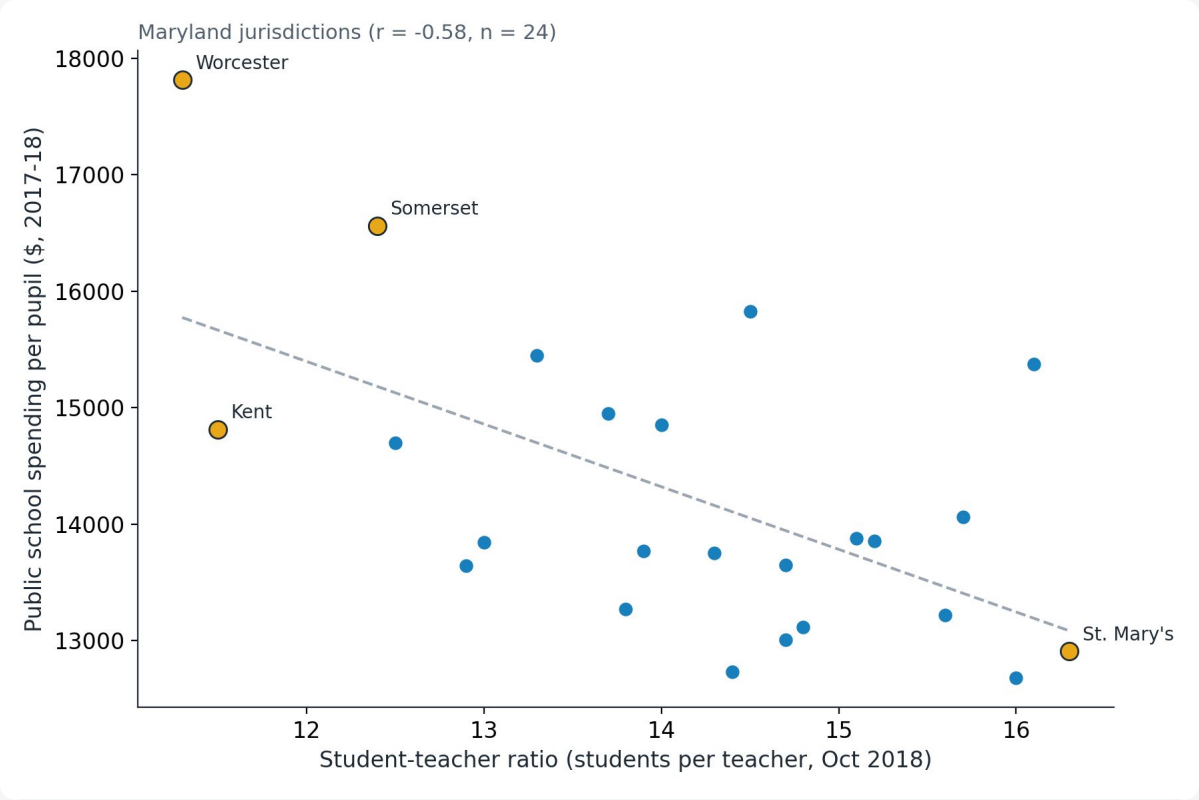
Howard; Montgomery is 59.0%

The next highest is Anne Arundel at 40.9%. The median is 31.4%, so these two sit about 18 points above everyone else.

Source: U.S. Census Bureau, 2014–2018

CHART 3 · SCATTER

Spending follows class size



$\bar{r} = -0.58$

class size vs. spending

Smaller classes cost more per pupil. Worcester has the fewest students per teacher, 11.3, and the highest spending, $17,814.

Source: Maryland State Department of Education, 2017–2018

WHAT COMES NEXT

Questions this data cannot answer yet

Spending over time

Compare spending when today's adults were in school with their degrees.

Cost of living

Adjust spending for local prices, since the same dollars buy different amounts in different counties.

More factors

Add household income and school results to see what sets Howard and Montgomery apart.

THE RESULT

Spending more per student does not go with more college graduates. Howard and Montgomery stand apart.

Different groups

Spending describes today’s students. Attainment describes adults 25 and older, schooled decades ago.

Only 24 counties

One or two outliers can move a correlation a lot.